Tyre Demand Forecasting by Geographic Zone in the Swiss Market: An Ensemble Approach with Bayesian Optimisation and Conversational Agent

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***Abstract*—This paper presents a demand forecasting system for tyre distribution in the Swiss market, developed as part of a curricular internship at SDG Group for a confidential client in the automotive sector. Switzerland poses a particularly challenging forecasting environment due to its bimodal Alpine seasonality, high territorial heterogeneity across 26 cantons, and structural trends such as premiumisation and electrification of the vehicle fleet. The methodology follows the CRISP-DM framework across four phases: data preparation (filtering, anonymisation, and geographic enrichment via spatial joins), exploratory data analysis (temporal, geographic, and product dimensions), predictive modelling, and deployment. A model ensemble combining XGBoost with Bayesian optimisation, XGBoost with macroeconomic features, SARIMA, SARIMAX, exponential smoothing (ETS), and Ridge regression selects the optimal model per canton. Walk-forward cross-validation with three temporal splits yields a mean MAPE of 28.8% for high-volume cantons, representing a 42% improvement over the historical mean baseline (50.9%). Results are deployed via an interactive Tableau Public dashboard and a conversational agent built on the Claude API.**

***Index Terms*—Bayesian optimisation, conversational agent, demand forecasting, ensemble learning, exploratory data analysis, SARIMA, Switzerland, time series, tyres, XGBoost.**

I. Introduction

Demand forecasting is a core challenge in supply chain management. The ability to anticipate demand accurately has become a decisive competitive advantage, particularly when geographic heterogeneity and complex seasonality are present [1]. In the tyre distribution sector, demand is strongly conditioned by climatic factors, vehicle fleet evolution, and structural market trends such as electrification, premiumisation, and the growing share of SUVs.

The Swiss market is a particularly rich case study: its Alpine geography imposes a marked bimodal tyre-change seasonality, its 26 cantons exhibit very different socioeconomic and mobility profiles, and its vehicle fleet is undergoing rapid transformation. The central problem is to estimate monthly tyre sell-out volume per canton for a 12-month horizon, combining sell-in, sell-out, carpark, and market-potential datasets.

Machine learning approaches, especially gradient boosting methods, have shown competitive results in demand forecasting when paired with adequate feature engineering [2]. Hybrid ensemble strategies that combine statistical and ML models further improve accuracy, as demonstrated in the M4 competition [3]. Bayesian hyperparameter optimisation outperforms grid search in high-dimensional parameter spaces at lower computational cost [4].

This paper is structured as follows. Section II describes the problem and datasets. Section III outlines the methodology. Section IV presents results. Section V concludes.

II. Problem Analysis

A. Available Datasets

Five client datasets were provided by SDG Group. Table I summarises their main characteristics. Due to confidentiality constraints the client name is not disclosed.

TABLE I
Summary of Available Datasets

Dataset	Temporal	Geo.	Rows
Sell In	2023-2026 (monthly)	Postal code	294,132
Sell Out	2018-2026 (monthly)	Postal code	704,699
Customer Master	Static	CP + coords	3,464
Carpark	2026 (annual)	Postal code	3,963,645
Tire Potential	2024-2026 (monthly)	Postal code	5,069,750

Sell In records transactions between the manufacturer and its distributors, including date, ship-to customer ID, tyre type, season, brand tier, and quantity. **Sell Out** captures consumer-facing sales from six voluntarily reporting companies, covering the period 2018-2026, the longest historical series available and the primary target variable. **Customer Master** acts as a geographic bridge between Sell In and Swiss cantons via postal codes and GPS coordinates. **Carpark** provides the 2026 vehicle fleet composition per postal code, including EV share, vehicle age, and tyre size type. **Tire Potential** offers a monthly market-potential estimate per postal code, used to assess the relative penetration of the reporting companies.

B. Swiss Market Characteristics

Several structural features of the Swiss market condition both analysis and modelling:

- **Bimodal seasonality**: demand peaks in spring (summer-tyre change) and autumn (winter-tyre change), with pronounced valleys in between.
- **Sell In - Sell Out lag**: distributors stock up ahead of each season, creating an approximate two-month lead of Sell In over Sell Out. This makes Sell In a natural leading indicator.
- **Cantonal heterogeneity**: volume ranges from over 5,000 units/month in the largest cantons to fewer than 10 in the smallest, making a single global model suboptimal.
- **Partial market coverage**: Sell Out reflects only the six reporting companies, requiring contextualisation against Tire Potential.
- **Structural trends**: the fleet is shifting towards premium tyre segments and larger rim sizes (SUVisation), while EV adoption is growing.
- **Macroeconomic sensitivity**: the Swiss consumer price index (CPI) and year-on-year inflation affect average tyre price and purchasing behaviour.

C. Specific Challenges

Beyond the structural characteristics above, three operational challenges shaped the system design. First, a logistic outlier: postal code 4624 (Härkingen, Solothurn) concentrates 17.2% of total Sell Out volume due to a distribution centre, distorting geographic analysis. Second, data confidentiality: 17 sensitive columns across both files required anonymisation with consistent cross-dataset mappings before any analysis. Third, demand intermittency in low-volume cantons: Uri and AppenzellAusserrhoden average fewer than 60 units/month, making their demand essentially stochastic and impossible to forecast reliably with any statistical model [5].

III. Implementation

A. Data Preparation

Geographic enrichment: neither Sell In nor Sell Out contains a canton field directly. Sell Out postal codes were mapped to cantons via a lookup table (99.6% coverage). For Sell In, Customer Master was joined on `SHIP_TO_CUST_ID`, with a fallback to `PAYER_CUST_ID` when the ship-to address was missing, recovering coverage from 87.7% to 99.0% of records (99.4% of volume). Remaining unresolved postal codes were assigned via GeoPandas spatial join against the GADM 4.1 Swiss canton shapefile [7], achieving 99.1% coverage.

Anonymisation: 17 sensitive columns (11 in Sell In, 6 in Sell Out) were replaced with generic identifiers using consistent JSON mapping dictionaries stored outside the project repository.

B. Exploratory Data Analysis

EDA (Esploratory Data Analysis) was conducted along three dimensions. **Temporally**, Pearson correlations between temperature (Open-Meteo API, Zürich) and Sell Out confirmed a significant two-month lag: $r = 0.535$ for Winter tyres and $r = -0.446$ for Summer tyres (both $p < 0.001$). All-Season tyres showed no significant climate correlation, suggesting convenience-driven purchase behaviour. The mean Sell In - Sell Out gap was 24.1%, confirming advance stocking. **Geographically**, concentration curves showed that 14% of postal codes account for 80% of Sell In volume. The Sell Out / Tire Potential ratio identified Genève (42.7%), SanktGallen (54.6%), and Schaffhausen (29.3%) as the cantons with the greatest untapped growth potential. **At product level**, the premium brand tier grew from 28% to 80% of the mix between 2018 and 2026, while rim size RIM 16 fell from 37% to 22% as RIM 18 rose from 10% to 21%, both trends linked to SUVisation of the Swiss fleet.

C. Feature Engineering

A master dataframe at canton-month granularity integrates Sell Out (target variable), Sell In (leading indicator, available from 2023), and climate data. Two feature sets were constructed:

v1 - Base temporal and climate (15 features): month, quarter, cyclic month encoding (sin/cos), Sell Out lags (1, 2, 3, 6, 12 months), 3- and 6-month rolling means, and temperature/snow/precipitation with a two-month lag.

v2 - Macroeconomic enrichment (v1 + 7 features): Swiss CPI, year-on-year inflation (World Bank), total vehicles per canton, EV share, mean fleet age, premium tyre share, and winter tyre share (Carpark dataset).

D. Canton Stratification

The failure of a global XGBoost model (MAPE 92.5%) on low-volume cantons motivated stratification by mean monthly volume over 2023-2026. Cantons with mean ≥ 500 units/month were labelled *large* (13 cantons); the rest were labelled *small* (13 cantons).

E. Model Families

Six model families were evaluated:

1) **XGBoost v1**: gradient boosting on feature set v1, per-canton for large cantons, global for small ones. Initial MAPE: 38.8% (large).
2) **XGBoost v2**: feature set v2. MAPE: 32.6% (large, -16% vs. v1).
3) **XGBoost Bayes**: feature set v2 with Optuna Bayesian optimisation (100 trials, TPE algorithm). MAPE: 29.3% (large, -25% vs. grid search).
4) **SARIMA** $(1,1,1)(1,1,1)_{12}$: fitted individually per canton. Competitive in cantons with very stable seasonality (e.g., Aargau: 13.9%).
5) **SARIMAX**: SARIMA extended with exogenous variables (CPI, inflation, EV share, temperature lag).

Adopted only for Vaud, where it improved MAPE by 7.0 pp.

6) **ETS / Ridge**: exponential smoothing and L2-regularised regression as fallbacks for cantons with insufficient data for complex models.

F. Validation Strategy

Walk-forward cross-validation with three temporal splits (December 2023, June 2024, December 2024) was used throughout. The mean MAPE over the three test periods provides a conservative but realistic estimate of production error, substantially more honest than a single train/test split (single-cut MAPE: 29.8% vs. walk-forward: 59.9% globally, reflecting high variability in small cantons).

G. Ensemble Construction

The final ensemble assigns each canton the model with the lowest walk-forward MAPE, following the best-per-subgroup selection strategy shown to be effective in heterogeneous demand settings [3].

H. Deployment

An interactive **Tableau Public** dashboard integrates five views (time series, prediction map, product mix, seasonality by type, and canton-level MAPE) with globally synchronised canton and model filters. A **conversational agent** built on Flask and the Anthropic Claude API (`claude-sonnet-4-20250514`) embeds the Tableau dashboard and provides natural-language interpretation of forecasts. The system prompt encapsulates the full project context, canton-level results, reliability criteria, and available Plotly charts, enabling the agent to classify forecast reliability, generate on-demand visualisations, and contextualise results without retraining.

IV. RESULTS

A. Exploratory Data Analysis

The bimodal seasonality pattern proved highly stable across all four years with complete data (2022-2025), with normalised Sell Out curves almost identical year-on-year (Fig. 1). The two-month temperature lag confirmed a systematic advance-stocking behaviour: distributors respond to expected climate changes rather than current conditions. The mean Sell In - Sell Out gap (24.1%) showed a declining trend from 2024, suggesting progressive inventory optimisation by distributors.

Geographic analysis confirmed that Härkingen (Solothurn, CP 4624) is a logistic outlier concentrating 17.2% of national Sell Out, consistent with it hosting a major distribution centre. Excluding this canton from geographic visualisations while retaining it in cantonal modelling was the adopted approach.

Premium tyre share grew from 28% to 80% between 2018 and 2026 (surpassing Quality in 2019), while the Budget segment virtually disappeared. RIM 18 doubled its share. These structural shifts have direct implications for revenue per unit and inventory planning.

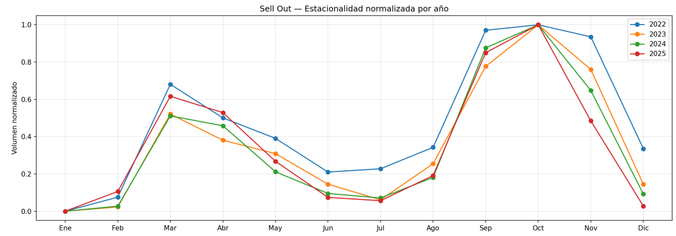

Fig. 1. Normalised Sell Out seasonality by year (2022-2025). All four years show an almost identical bimodal pattern, confirming the stability of spring and autumn demand peaks.

B. Predictive Modelling

Table II compares all evaluated models via walk-forward MAPE.

TABLE II
WALK-FORWARD MAPE COMPARISON BY MODEL

Model	Mean MAPE	Scope
Baseline: Naive	66.4%	Global
Baseline: Historical mean	50.9%	Global
XGBoost v1 (Grid Search)	39.0%	Large cantons
SARIMA	42.2%	Large cantons
ETS	≈40.0%	All cantons
XGBoost v2 (macro features)	32.6%	Large cantons
XGBoost Bayes (Optuna)	29.3%	Large cantons
Final ensemble (WF)	**59.9%**	**Global (26 cantons)**
Large cantons only	28.8%	13 cantons
Excl. 2 unpredictable	40.1%	24 cantons

The final ensemble achieves a mean walk-forward MAPE of 28.8% on the 13 large cantons, representing a 42% improvement over the historical mean baseline (50.9%). This places the results within the "good" range reported in industrial demand forecasting literature ($< 30\%$) [5]. Bayesian optimisation (Optuna) outperformed grid search by 25% (29.3% vs. 39.0%), consistent with findings in AutoML research [8].

Two cantons proved structurally unpredictable: Uri (MAPE 252.1%, $\sigma = 247.1$ pp) and AppenzellAusserrhoden (MAPE 332.3%, $\sigma = 367.9$ pp). Both have mean monthly volumes below 60 units and standard deviations exceeding the mean, making any historical-pattern-based model fundamentally inadequate [5]. Together they represent less than 0.4% of Swiss national volume. Excluding them, 20 of 24 cantons (83%) fall below the 50% MAPE acceptability threshold.

Table III shows the 12-month demand forecast (March 2026 - February 2027) for the six largest cantons.

XGBoost with Bayesian optimisation was assigned to 8 of 13 large cantons (62%) and to 8 of 13 small cantons (62%) as XGBoost v2, making it the dominant model family (46% + 31% = 77% of cantons). SARIMA and SARIMAX were assigned to cantons with the most stable and pronounced seasonality.

TABLE III
12-MONTH DEMAND FORECASTS FOR SELECTED CANTONS

Canton	Model	Units	MAPE (WF)
Solothurn[a]	XGB_Bayes	64,782	27.3%
Zürich	XGB_Bayes	49,346	26.4%
Vaud	SARIMAX	30,632	20.6%
Bern	XGB_Bayes	26,983	30.1%
Ticino	XGB_Bayes	23,660	18.7%
Aargau	XGB_Bayes	23,582	15.0%
Switzerland total	Ensemble	320,309	,

[a]Inflated by the Härkingen logistic centre.

C. Dashboard and Conversational Agent

The Tableau Public dashboard integrates five views: (1) historical Sell Out plus 12-month predictions per canton with a vertical reference line at the forecast start; (2) a bubble map with circles proportional to predicted demand; (3) stacked 100% bars showing brand tier mix evolution; (4) seasonality curves by tyre type; and (5) horizontal bars of walk-forward MAPE per canton, colour-coded by assigned model. A global canton parameter synchronises all views regardless of source dataset, solving Tableau's native filter limitation across multiple data sources.

The conversational agent demonstrated four capabilities in testing: automatic reliability classification per canton (Very Reliable / Acceptable / Weak / Unpredictable) based on MAPE thresholds embedded in the system prompt; executive summaries of global results in non-technical language; contextual chart recommendation and automatic Plotly generation; and direct deep-link access to the Tableau dashboard for each response. Mean response latency in live demonstrations was under 10 seconds, meeting the stated non-functional requirement.

V. CONCLUSIONS

This paper presented a complete tyre demand forecasting system for the Swiss market, combining rigorous data preparation, multidimensional exploratory analysis, an ensemble of six model families, and a dual deployment layer (interactive dashboard and conversational agent). The main conclusions are as follows.

Bimodal seasonality is the dominant, highly stable pattern. Spring and autumn peaks correlate significantly with temperature at a two-month lag, providing a reliable seasonal signal for all models.

No single model is universally optimal. Cantonal heterogeneity, spanning three orders of magnitude in monthly volume, requires per-canton model selection. XGBoost with Bayesian optimisation dominated large cantons; SARIMA proved more accurate in cantons with very pure seasonality; ETS and Ridge were the best available options for low-volume cantons.

Bayesian optimisation significantly outperforms grid search. A 25% MAPE reduction (29.3% vs. 39.0%) was achieved with Optuna at lower computational cost per evaluation, supporting its adoption in territorial demand forecasting projects.

Macroeconomic and fleet variables improve accuracy. Incorporating CPI, inflation, EV share, and fleet size reduced the XGBoost MAPE by 16% (38.8% to 32.6%), suggesting that structural market context is a significant predictor beyond historical sales patterns alone.

Walk-forward validation is essential for honest evaluation. The single-cut MAPE (29.8%) substantially underestimates the production error compared to the walk-forward estimate (59.9% globally, 28.8% for large cantons), with the difference driven entirely by small-volume cantonal volatility.

Low-volume cantons are structurally unpredictable. Uri and AppenzellAusserrhoden should be managed via fixed safety stock or reactive replenishment rather than statistical forecasting, a recommendation the conversational agent communicates automatically when queried about these cantons.

Future work should explore Prophet for multi-seasonality modelling, LSTM or Transformer architectures for sequence learning (subject to dataset size constraints), monthly matriculation data to capture fleet dynamics temporally, postal-code-level forecasting for the top 300 codes, and a tool-calling extension of the conversational agent enabling real-time queries against live data.

REFERENCES

[1] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "The M4 competition: Results, findings, conclusion and way forward," *Int. J. Forecasting*, vol. 34, no. 4, pp. 802-808, 2018.

[2] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, pp. 785-794, 2016.

[3] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "The M4 competition: 100,000 time series and 61 forecasting methods," *Int. J. Forecasting*, vol. 36, no. 1, pp. 54-74, 2020.

[4] J. Snoek, H. Larochelle, and R. P. Adams, "Practical Bayesian optimization of machine learning algorithms," *Advances in Neural Information Processing Systems*, vol. 25, pp. 2951-2959, 2012.

[5] A. A. Syntetos and J. E. Boylan, "The accuracy of intermittent demand estimates," *Int. J. Forecasting*, vol. 21, no. 2, pp. 303-314, 2005.

[6] R. Wirth and J. Hipp, "CRISP-DM: Towards a standard process model for data mining," in *Proc. 4th Int. Conf. Practical Applications of Knowledge Discovery and Data Mining*, pp. 29-39, 2000.

[7] K. Jordahl et al., "GeoPandas: Python tools for geographic data," *J. Open Source Software*, vol. 6, no. 65, p. 3424, 2021.

[8] M. Feurer, A. Klein, K. Eggensperger, J. T. Springenberg, M. Blum, and F. Hutter, "Efficient and robust automated machine learning," *Advances in Neural Information Processing Systems*, vol. 28, 2015.

[9] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proc. 25th ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, pp. 2623-2631, 2019.

[10] G. E. P. Box, G. M. Jenkins, G. C. Reinsel, and G. M. Ljung, *Time Series Analysis: Forecasting and Control*, 5th ed. Wiley, 2015.

[11] C. W. Chase, *Next Generation Demand Management: People, Process, Analytics, and Technology*. Wiley, 2013.

[12] Anthropic, "The Claude 3 model family: Opus, Sonnet, Haiku," 2024. [Online]. Available: `https://www.anthropic.com/news/claude-3-family`